

Data Cleaning as a Process

❑ Data discrepancy detection

- ❑ Use metadata (e.g., domain, range, dependency, distribution)
- ❑ Check field overloading
- ❑ Check uniqueness rule, consecutive rule and null rule
- ❑ Use commercial tools
 - ❑ Data scrubbing: use simple domain knowledge (e.g., postal code, spell-check) to detect errors and make corrections
 - ❑ Data auditing: by analyzing data to discover rules and relationship to detect violators (e.g., correlation and clustering to find outliers)

❑ Data migration and integration

- ❑ Data migration tools: allow transformations to be specified
- ❑ ETL (Extraction/Transformation/Loading) tools: allow users to specify transformations through a graphical user interface

❑ Integration of the two processes

- ❑ Iterative and interactive (e.g., Potter's Wheels)